# Understanding the earthquake research

A plain language guide to the questions, findings and limits

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This edition adds an executed background-rate control and audits the saved simulation histories, fitted parameters and mathematical assumptions. Original inputs, calculations and reproduction records remain available; unsupported historical results remain withdrawn.

Can a model learn when an earthquake makes later activity less likely, as well as when it makes later activity more likely? And does that extra detail actually improve forecasts?

That is the central question of this research. An earthquake catalogue records events: when they happened, where they were located and their reported magnitudes. It does not directly label how much each event encouraged or discouraged later earthquakes. The research asks what can be learned from those records, what requires extra assumptions, and what survives a predictive test.

## The main finding

In the completed observed-catalogue comparisons, adding the tested inhibition mechanisms has not established better predictive performance than the corresponding models without inhibition. Some controlled simulations do show gains. This is a finding about the tested models and studies, not proof that physical inhibition never happens.

## Three different achievements

A model can be mathematically well defined, have parameters that could in principle be distinguished, and still fail to improve predictions from a finite catalogue. The manuscript treats these as separate questions. A good fit to the past is not enough to settle any of them.

The contribution is therefore broader than a league table of forecasts. It develops restricted mathematical results, tests where learning becomes difficult, and records both the successes and failures of particular forecasting procedures.

## How to read this guide

Pages 2–4 explain the models and the tests. Pages 5–6 explain the completed simulation and regional findings. Pages 7–8 set out the remaining limits, define common terms and point to the relevant manuscript chapters. No equations are needed.

These are research comparisons using historical catalogues and simulations. The project has not established an operational earthquake forecasting service.

02 The ingredients

# What goes into a model

Here, a model is a recipe for assigning a changing rate of earthquake occurrence across time and, in the spatial studies, across a region. It gives a distribution of possible activity rather than a single certain date and location for the next earthquake.

### Background

Activity not assigned to earlier recorded events

### Triggering

Earlier events raise the modelled rate

### Inhibition

A proposed reduction in part of the rate

Different models combine these ingredients in different ways.

## Background activity

This is the part of the model's event rate that is not attributed to triggering by recorded earlier events. It may change over time and place. Calling an event a background event is a modelling description; it is not a direct observation of its geological cause.

## Triggering and inhibition

Triggering raises the modelled rate after earlier events. Inhibition lowers a specified part of that rate. In the original local-inhibition model, only the background contribution is reduced. One newer version also reduces the triggered contribution. Those are different hypotheses, even when they use the same event history.

## The place and the size of an event matter

The spatial models distinguish a region's background pattern from the pattern of later events around an earlier earthquake. A map of long-term activity and a model of an aftershock cloud do different jobs. The research tests choices about both.

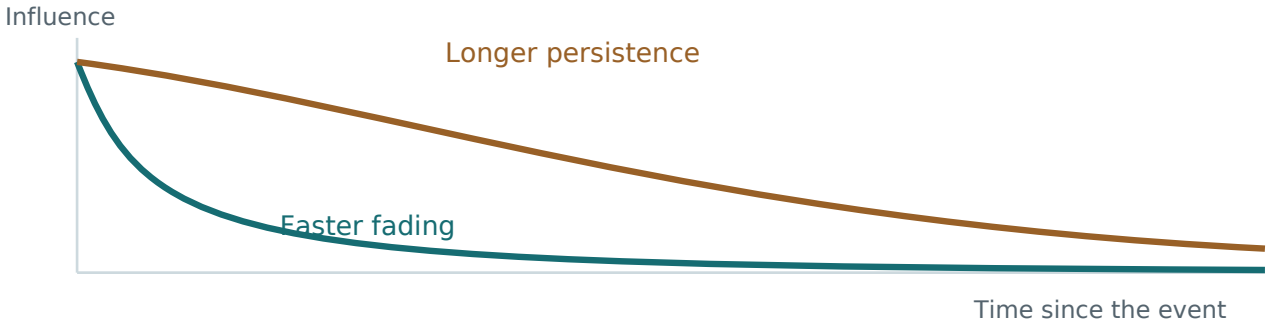
The original local-inhibition model gives all eligible past events the same suppressing weight. LI+, the revised family, gives larger magnitudes more weight. This changes how event size enters the hypothesis. It does not by itself make the model more accurate or turn it into a measured map of stress inside the Earth.

A separately executed constant-background fit links the stability model to the data. It uses the same intensity family as an earlier sensitivity comparison, with a different fitting recipe. Inhibition loses in the early and late periods and is switched off in the middle. The theorem does not establish practical equilibration over the catalogue period.

03 The mathematical questions

# Memory and what can be learned

Memory means that the influence of an earlier event fades over time. A short memory concentrates influence soon after an event. A longer memory allows it to persist. It is a feature of the model, not an extra quantity directly recorded in the catalogue.




Schematic only. These are not fitted curves or earthquake observations.

## Many simple fading effects can represent a complex memory

Some computations combine a finite collection of simple fading components, each with its own speed. This gives the computer a compact state to update as events arrive. It can approximate a more complicated memory law over a chosen range, but a finite approximation is not the same mathematical object as an exact long-memory law.

## Stability has a precise and limited meaning

For the physical finite-mode model, the results require constant loading, fewer than one directly triggered event per event on average, and suitable conditions on the response and event properties. Under those assumptions, the theory controls expected activity. That does not mean every simulated sequence is small or that all measures of variability are finite. The manuscript shows why first-moment results—results about averages—can coexist with difficulties involving second moments, which concern variability.

## Some settings cannot be distinguished

Imagine turning the strength of a model's inhibitory effect to zero. Changing its proposed range or duration then makes no difference to the forecast. The catalogue cannot identify those unused settings from that forecast. A reported range at zero strength is a convention, not a discovered physical length scale.

Even away from zero, two effects can compete to explain the same pattern. A changing background can make it harder to learn an inhibition effect. Knowing that parameters are distinguishable in an ideal mathematical law does not guarantee accurate estimates from the particular catalogue available.

04 Fit, prediction and calibration

# How the comparisons work

## Fit first, then test on later observations

The model's adjustable settings are fitted using a training history. A predictive comparison then scores observations outside that fitting period under the declared rules. The relevant comparison uses the same events, time window and region for both models.




Illustrative split. Window lengths and study designs vary.

## A score rewards useful probabilities

The research uses log scores: a way of evaluating the probabilities or rates a model assigns to what actually happens. A model cannot win simply by announcing high activity everywhere. Its score also accounts for the activity it predicted across the full observation window. Higher scores are better within the same scoring comparison.

Scores can be separated into contributions for timing, location and magnitude. A model may improve where it places activity while making the timing worse. The overall result and the component results therefore need to be read together.

## Calibration asks a different question

A forecast that assigns a one-in-five chance to an event should, across suitable repeated comparisons, see that event about one time in five. That is the idea of calibration. In this research, simulated futures are compared with the actual future to look for discrepancies in counts, event spacing and locations.

### Winning does not mean passing every test

One model can score better than another while both give poor descriptions of important features of the data. The benchmark's score advantage is not a calibration certificate.

A log-score difference reported in nats is a difference on a logarithmic scale. It is not a percentage improvement or a number of earthquakes correctly predicted.

05 Experiments with known rules

# What the controlled studies show

Simulations create catalogues from specified rules. The studies below include recovered historical experiments and a new comparison on their saved histories.

## Knowing the background can change the answer

The historical study compares supplied laws and estimated background parameters on 1,000 no-inhibition histories and 150 alternatives. Detection falls when the background must be estimated. Each procedure uses its own reference bank from the known generating law; this is not calibration for an unknown real-world law.

## A nonzero fitted effect is not enough

The magnitude-history study uses 100 paired simulation units at three levels of magnitude dependence. Matching how magnitude affects triggering reduces apparent evidence for inhibition, but small positive fitted effects remain common. Same-bank thresholds are descriptive checks, not independent false-alarm validation.

## The expanded inhibition controls have mixed outcomes

In the original 80-history local controls, background-only inhibition gains on average under broad, long-lasting generation; total attenuation loses. The narrow/short histories reject no proposed background events and follow the same paths as uninhibited histories using the same random streams and settings. The generating rules differ, but these scores do not test recovery of realized suppression. Penalties choose no inhibition in both null settings and the narrow/short setting. The shared background rate is fixed.

### Which law generated those histories matters

The matched extension generates magnitude-weighted background-only and total inhibition. In its narrow/short immigrant setting, 19 of 20 histories reject no events. A positive generating strength did not create an informative realized signal.

A new control fits the uninhibited background rate on training data for all 80 matched-law histories. Matched broad/long mean gains fall from 48.39 to 5.16 nats for background-only inhibition and from 53.23 to 9.74 for total inhibition. Wins fall from 20/20 to 19/20 and 18/20. The original inhibited fits and gates stay unchanged. Gains survive this intercept-only check but depend strongly on the comparator.

Parameter summaries now expose boundary selections. Narrow/short raw losses remain. These controls do not establish parameter recovery or compare jointly fitted nuisance parameters; ideal null score moments can diverge.

06 New Zealand, California and Japan

# What the regional studies show

## New Zealand is a development record

The same held-out earthquakes were examined across successive modelling studies. This helped develop and diagnose the models, but it makes the later comparisons exploratory. Freezing the next procedure does not make previously inspected observations new again.

Moving from a coarse location reference to a continuous spatial model can produce large score gains. That shows a benefit over the particular coarse reference. It does not establish superiority over a strong competing spatial model.

## California separates the main comparison from the backtest

The inverse-power aftershock benchmark, EP, beats OC in each original primary California test fold, chiefly through location. Original inhibition and magnitude-weighted background-only inhibition select zero strength in all three observed folds. The revised model that also suppresses triggered activity loses held-out score in every fold; its complexity penalty selects no inhibition.

A separate episode-based backtest now covers five eligible historical episodes. It fits each model using only the history before that episode, but the roster was selected retrospectively. Results vary by episode: EP does not win every one. These episodes are not five independent confirmations.

## Japan exposes a timing problem

Against the reference model with a fixed triggering-time law, the timing gain seen in the original California comparison becomes a loss in Japan. Fixing its background trend to be constant reduces the overall timing deficit but does not remove it. OC also remains behind EP jointly in that study. The trend choice is therefore not a complete explanation of the reversal.

### A better location model does not settle the whole forecast

The completed spatial ablations retain an overall EP advantage over OC in California and New Zealand, mainly through location. Individual fold results vary, and several model features change across the arms. The comparison does not isolate a single physical cause.

A fold is one training-and-test comparison. Different studies use different windows and model settings, so their scores should not be treated as one interchangeable league table.

07 The boundaries of the evidence

# Interpreting the evidence

## A failed model does not rule out a physical process

A model can lose because its form is too restrictive, its background is inadequate, its fitting search misses better settings, the catalogue contains too little information, or the proposed effect adds little predictive value in that setting. These studies narrow some explanations. They do not identify a unique geological cause or prove the absence of physical inhibition.

## Looking back is different from issuing a forecast

Using only earlier data when fitting is an important safeguard. It does not turn a retrospectively chosen set of episodes into a genuinely prospective study. Overlapping training histories and related earthquake sequences also limit how many independent pieces of evidence the comparisons supply.

## Readers can now rebuild the study inputs

Original provider records and selection rules are now available for the New Zealand reconstruction, California and Japan. Replays rebuild their study catalogues from those records. A fresh run of the frozen New Zealand comparison also reproduced its fitted parameters, event-level scores and numerical controls exactly. This makes those calculations inspectable and repeatable; reused observations remain part of model development.

## Better scores do not settle calibration

The completed study compares OC and EP with 999 simulated futures at each of three forecast origins. Combined checks using continuous timing and spatial statistics flag discrepancies for both models in every origin. EP can score better while still giving an inadequate description of some features. These checks hold fitted parameters fixed; they do not include uncertainty about those parameters.

## What the supplied records allow

The supplied histories, fitted states and code let readers inspect the completed comparisons. Original California checks covered every simulated path for the declared score fields, with separate complete-vector checks on the selected subset. Later packaging tests are identified separately from those original executions.

The F continuation has been removed because required original fitted states could not be recovered. Its older sources and numerical records remain archived; the supported earlier nuisance study is retained.

08 Finding your way through the manuscript

# A small glossary and reading map

| | |
|---|---|
| **Catalogue** | The recorded times, locations and magnitudes of selected earthquakes. |
| **Parameter** | An adjustable model setting, such as the strength or duration of an effect. |
| **Inhibition** | A mechanism that reduces a specified part of the modelled event rate. |
| **Identifiability** | Whether different settings could be distinguished from the model's observable law, in principle. |
| **Null control** | A simulated case without the effect the procedure is supposed to detect. |
| **Oracle calibration** | A simulation-based reference using the known generating law. |
| **Held-out data** | Observations outside a particular fitting step; repeated inspection can still make them part of model development. |
| **Zero-strength boundary** | The setting where inhibition is switched off and its unused range and memory cannot be identified. |
| **Calibration** | Agreement between a forecast's probability statements and repeated outcomes under the declared comparison. |
| **OC, EP, LI and LI+** | OC uses a regional background map to weight where triggered activity is placed; EP omits that weighting and uses its own distance-decay rule. LI adds local background suppression to OC; LI+ gives larger events stronger suppressing weights. |

## Where to read more

| | |
|---|---|
| **The model and its mathematical limits** | Chapters 2–5 |
| **Study design and controlled detection** | Chapters 6–7 |
| **Historical temporal and magnitude comparisons** | Chapters 8–9 |
| **Spatial models, regional transfer and backtests** | Chapters 10–11 |
| **Calibration and local inhibition** | Chapters 12–13 |
| **Overall interpretation and remaining questions** | Chapter 14 |

Read each result with its comparator, observations and assumptions. For source records, selection rules and runnable checks, see the repository's data guide (docs/data.md) and replay guide (reproduction/README.md). Readers can inspect the history, report errors and propose revisions; the manuscript and evidence records supply the detailed definitions.